n = 19 tau6 native per-shift solve, protocol v3 (frozen; Sage smoke gate + repeated alternating pairs)

2026-10-20

n = 19 tau6 native per-shift solve, protocol v3

Status (2026-10-20): frozen before any timing. The run is launched, and no result exists yet. v3 replaces v2. v2's Tier 1 run, 39376405326, was invalid: S0 on 2,682 systems, O1 and the field check passed, but every timed IC row (B, N, Naa and NX on all 16 points) crashed after its interval in `arm.sage`'s `relation_digests`. v1's and v2's protocols and archives are unchanged. The workflow is `n19-native-tau6-v3.yml`.

Root cause of v2's crash

See `diagnosis/README.md` for the full account. In short, the Sage preparser turns the literal `0` in `row.get(col, 0)` into a Sage `Integer`. Every value of the relation-rows digest was therefore an `Integer`, and `json.dumps` refused it: `TypeError: Object of type Integer is not JSON serializable`.

- The run's traceback places the failure at `h(rows)`, after the `ordered` and `canonical` digests succeeded. This holds in all 64 IC logs.
- A local emulation on v2 point p00 (`diagnosis/emulate_preparser.py`) runs v2's own function. In plain Python it succeeds. With integer literals made `Integer`, it fails exactly as on the runner.
- The smoke job reruns v2's `arm.sage` on rows p00-N and p00-B in the pinned image (`diagnosis/digest_repro.sage`).
- v2's local checks were pure Python, and no Sage check before the timed rows ever wrote an IC row.

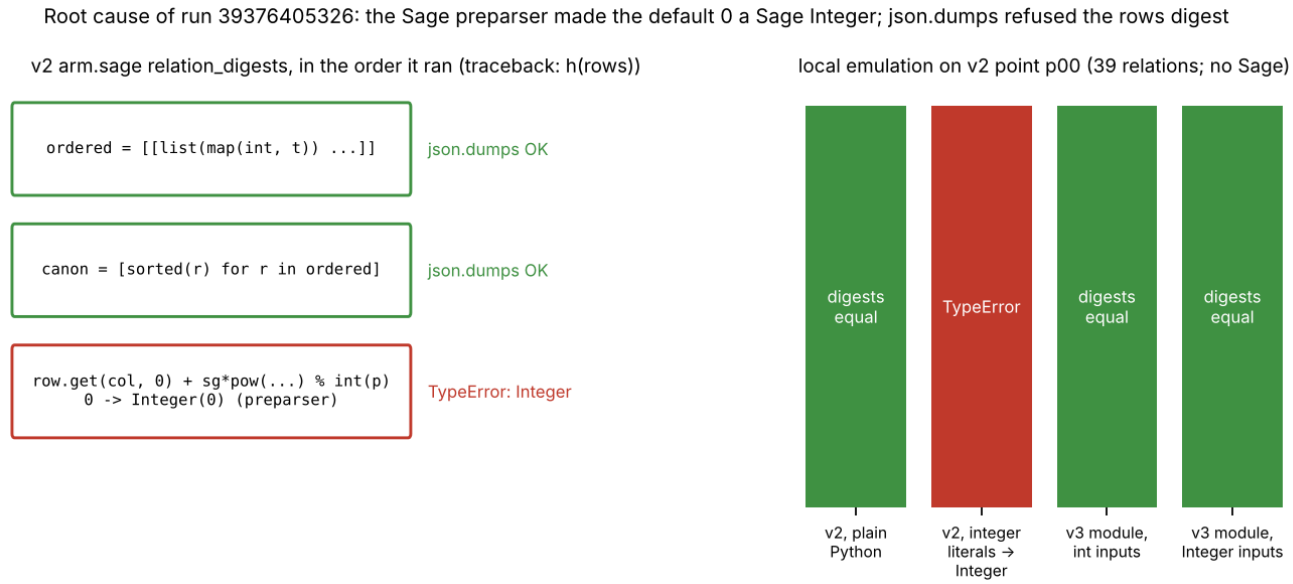


Figure 1: Root cause. Source: make_figures.py; vector: figures/crash_diagnosis.svg.

What v3 changes

	v2	v3
relation digests	in `arm.sage` (preparsed)	`relation_digest.py`, plain Python, `operator.index` on every input; same definitions
row serialization	`json.dump(..., default=jplain)`, falls back to `str()`	`relation_digest.plain`, strict: Sage Integer → int, reals → float, anything else raises
raw record	row file only	row bytes also on stdout as `raw_b64` + SHA-256; the verifier checks them against the row file and the schedule
relations	digests only	each IC row and O1 instance records its raw relations, and the verifier recomputes every digest
Sage smoke gate	none (the local smoke ran pure Python)	**job `sage-smoke`, untimed**: S0, O1, and v2 points p00 and p01 through B, N, Naa, NX and rho in the pinned image; 20 gates; the timed job `needs` it and uses its library
O1	counts and logs	also the relation digests of all ten archived ECDLPs (the crashing path, before any timed row), and the archived λ
S0	7 v1 systems + 300 + 2,375 multi = 2,682	+ v2's 300 re-derived from v2's seed + 300 new (v3 seed) = 2,982; the detail digest is recomputed
design	1 process per arm per point, 16 points	**20 points × (3 alternating B/N pairs + 3 N/Naa pairs + 1 NX/N pair + 1 rho)**, seeded ABAB/BABA starts, shuffled block order
interval	one-stage bootstrap over points (numpy)	**two-stage paired bootstrap** (points, then pairs), 10,000 draws, `random.Random(2026102003)`
isolation records	affinity only	per run: `Cpus_allowed_list`, the arm's affinity, `/proc/schedstat`, `/proc/stat`, `getrusage`, `isolated_bench` counters and noise, library digest before and after; `host_cpu_record` at start and end
verifier	gate names free-form	fixed gate set, equal to `protocol.json gates_required`; a missing, unknown or non-boolean gate makes the verdict invalid; 12 planted-corruption tests
trigger	dispatch (main); path-limited push (branch)	dispatch, and push limited to branch `claude/n19-native-v3` and this experiment's frozen paths (owner-approved pattern of v2)

The solve (`native/src/lib.rs`), the binding (`native_binding.py`), the question, the arms, the intervals (setup, online, cold) and the field check are v2's, byte for byte where they are code.

Frozen design

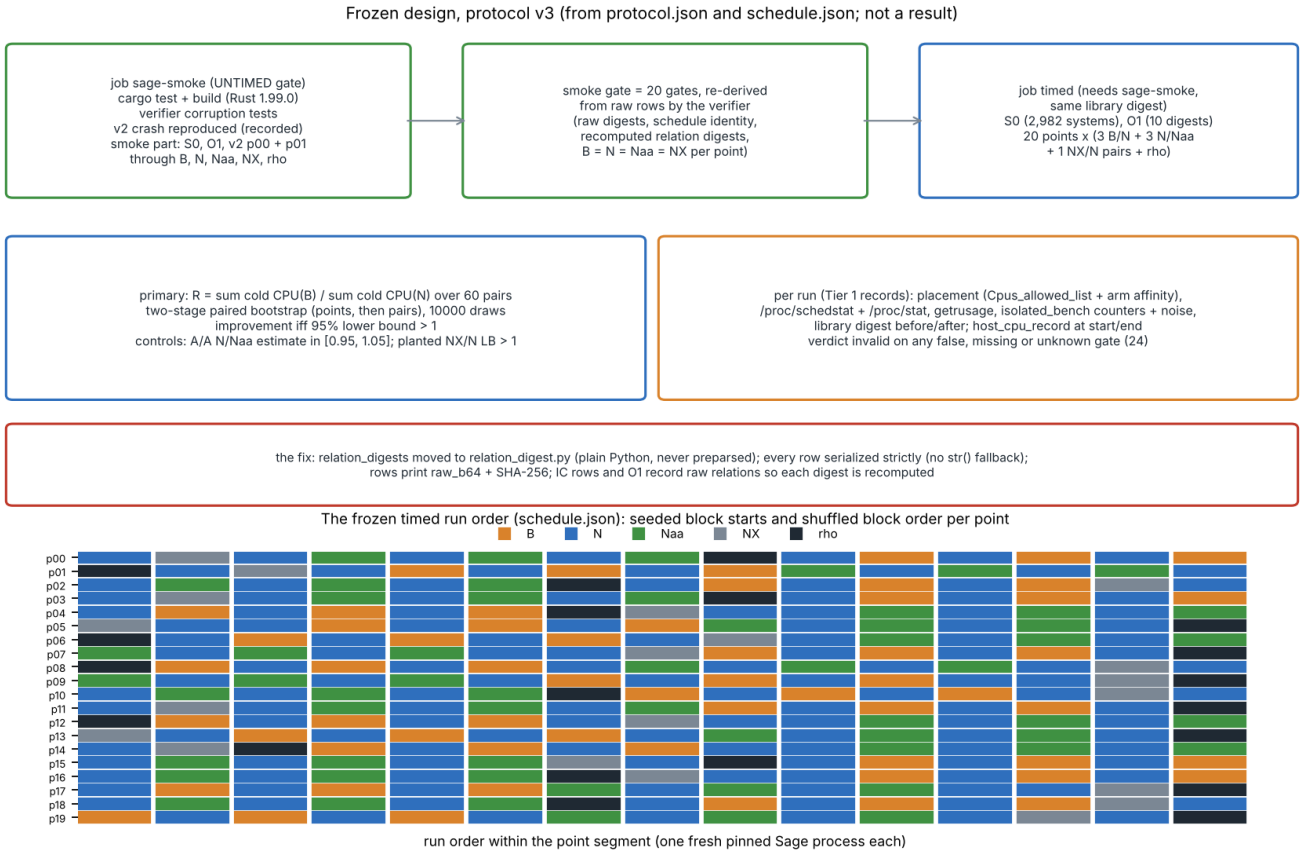


Figure 2: Frozen v3 design and run order (not a result). Source: make_figures.py; vector: figures/design.svg.

- **Primary.** R = Σ cold CPU(B) / Σ cold CPU(N) over the 60 primary pairs, where cold = setup + online, `time.process_time()` inside each fresh pinned Sage process. The interval is a two-stage paired bootstrap at 95%. Improvement iff the lower bound is above 1, regression iff the upper bound is below 1, otherwise not separated.
- **Controls.**
  - A/A: N/Naa over 60 pairs, estimate inside [0.95, 1.05].
  - Planted: NX/N over 20 pairs, lower bound above 1. NX redoes the Sage solve on every shift system and compares it with the native one.
- **Gates.** There are 24 (20 in smoke):
  - schedule identity, raw digests, recomputed relation digests and one library digest;
  - pinning, CPU identity and the Tier 1 counters;
  - S0, O1 and the field check in every native process;
  - identical counts, logs, rows, canonical and ordered decompositions across every IC process of a point;
  - NX comparing every call, the archived generator, rho 64/64 and the warm batch;
  - the smoke gate and complete pairs.

  Every gate is listed in `protocol.json gates`.
- **Panel.** 20 new scalars (`inputs/panel.json`, role `n19-native-heldout-v3`, seed `volcano-n19-native-tau6-v3-20261020`) None is equal to, the negative of, or a Frobenius image of an archived, v1 or v2 panel scalar. The order, S0, power and bootstrap seeds are all new.
- **Power** (`analysis/power.json`).
  - Model: per-process log-normal noise $\sigma = 0.10$ or 0.20, and per-point costs from v2's B processes or from the archived rows.
  - Power at a true ratio of 1.20 is at least 0.98. At 1.10 it is 0.995 ($\sigma = 0.10$) or 0.52 ($\sigma = 0.20$).
  - The rejection rate with no effect is at most 0.015.

  - The A/A window fails with no effect in 1–2% of panels at $\sigma = 0.10$ and 18–23% at $\sigma = 0.20$. v2's N launcher walls varied from 1.8 to 2.2 s, which points to the lower σ.
  - The expected effect is far larger: in v2's run, B processes took 57–99 s and N processes 1.8–2.2 s, Sage start-up included.
- **Runtime and time guard.**
  - The smoke job takes about 10 min. The timed job takes about 2 h, and about 2.6 h if every B process takes 100 s.
  - Before each segment, the guard skips it if the elapsed time plus max(its upper estimate, 1.1 × the longest point segment so far) exceeds 175 min. A skip makes the verdict invalid.
  - The job timeout is 200 min.
- **Tier.** Tier 1 as AGENTS.md defines it. The run uses one fresh GitHub-hosted ubuntu-24.04 runner and one pinned exposed CPU, and the image is `sagemath/sagemath:10.10@sha256:310dfe23...`. It is CPU-only, has one host class and is not replayed. A container affinity mask records placement, not host-wide exclusivity.

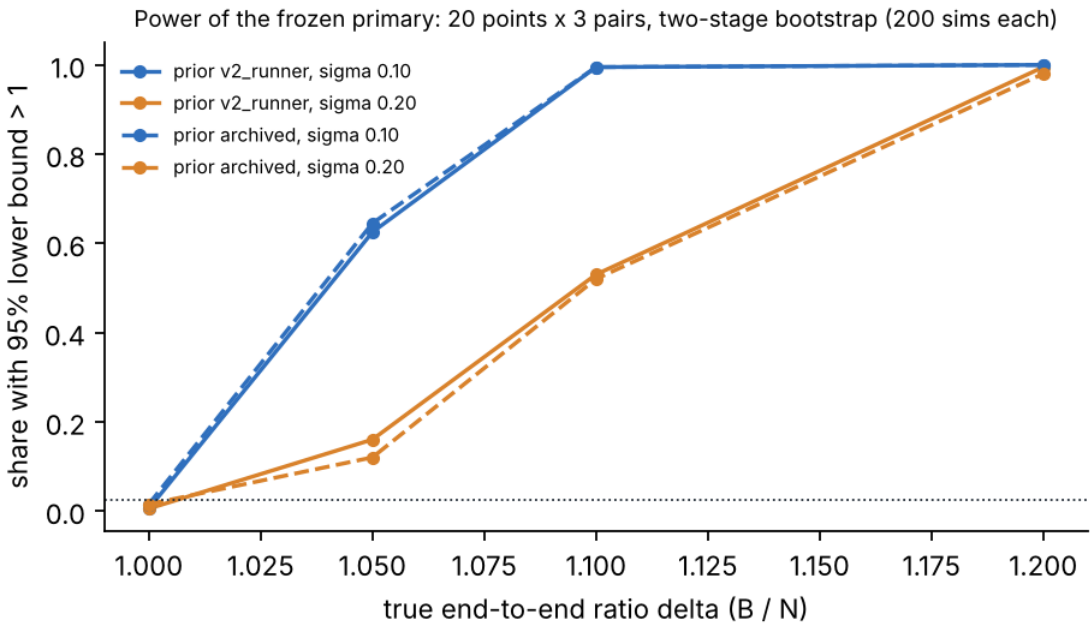


Figure 3: Power of the frozen primary (simulated). Vector: figures/power.svg.

Local checks before freezing (cloud-agent VM, no Sage; not timing evidence)

- **`cargo test --release --locked`.** 7 of 7 pass (rustc 1.97.0 locally; the runner pins 1.99.0).
- `smoke/local_v3.py` runs a pure-Python tauDlp with the v3 binding and digests (`smoke/local_v3.json`).
  - The ten archived scalars give counts and logs equal to run 38520305068.
  - All logs verify on the two smoke points and the 20 v3 points.
  - On the 20 v3 points, gb_calls average 5,446 (range 3,673 to 7,095); v2's panel averaged 5,653.
- `diagnosis/emulate_preparser.py` reproduces v2's error and shows that v3's digests are equal and type-safe.
- **`tests/test_verify_receipt.py`.** 12 of 12 pass on synthetic receipts:
  - a clean receipt passes;
  - each of these fails closed: a tampered row file, a digest that does not recompute, a schedule mismatch, a command mismatch, a raw record of another point, an unpinned process, an unknown, missing or non-boolean gate, a protocol gate list that differs, a failed A/A, counts that differ between arms, a time-guard skip, and a missing smoke receipt.
- `smoke/dryrun_fake_sage.py` checks the mechanics of the runner and verifier, with a stand-in `sage` that emits synthetic records (`smoke/dryrun_fake_sage.json`).
  - The real `run_schedule.py` launches every smoke run pinned and records the isolation fields. The real verifier passes 19 of 20 smoke gates.

  - The exception is `tier1_counters_recorded`: this VM has no `/proc/schedstat`. GitHub runners have it (n = 83 run 39359922687 records it).

Launch

The workflow is `.github/workflows/n19-native-tau6-v3.yml`.

- **Dispatch was tried first and refused.** `gh workflow run` returned 404, because the workflow is not on the default branch, and the GitHub integration returned 403 ("Resource not accessible by integration").
- **The push trigger launched it.** It is path-limited to this branch and this experiment's non-report paths, the owner-approved pattern from v2. The push of commit `92d85f14` started run 39384735073: first job `sage-smoke`, then job `timed`, which needs it.
- No result has been read. This README, the figures and the smoke and diagnosis files do not trigger a run.

Files

- **Code.**
  - `arm.sage` (one row per process), `relation_digest.py`, `native_binding.py` and `native/`;
  - `run_schedule.py` (the launcher and Tier 1 records), `verify_receipt.py` (fail closed) and `stats.py`.
- **Frozen inputs.**
  - `make_protocol.py` → `protocol.json`;
  - `make_panel.py` → `inputs/panel.json`;
  - `make_schedule.py` → `schedule.json`, with the parts smoke and timed;
  - `power_check.py` → `analysis/power.json`;
  - `inputs/multi_systems.json`, v2's 2,375 multi-solution systems.
- **Checks.** `tests/`, `smoke/` and `diagnosis/`.
- **Report.** `make_figures.py`, `figures/`, `README.pdf` and `report_header.tex`.